

TEACHER: ME

Year 11

July, 2011

General Instructions:

- Each question attempted is to be started on a NEW PAGE, clearly marked with the number of the question, your name and class on the top right hand side of the page
- Write using black or blue pen
- APPROVED CALCULATORS may be used
- All necessary working should be shown. Marks may be deducted if working is poorly set out or difficult to read.

[illegible]

Question 1 (8 marks)

- a) Evaluate $(\frac{1}{5})^{-1.2}$ correct to 3 significant figures. 2
- b) Find a and b such that $\frac{1}{\sqrt{5}-2} = a + b\sqrt{5}$ 2
- c) Write the equation of the circle with centre $(-3,4)$ and radius 6 units. 1
- d) Solve $|2x + 3| = 9$. 2
- e) Write the domain of the function $f(x) = \sqrt{6-x}$. 1

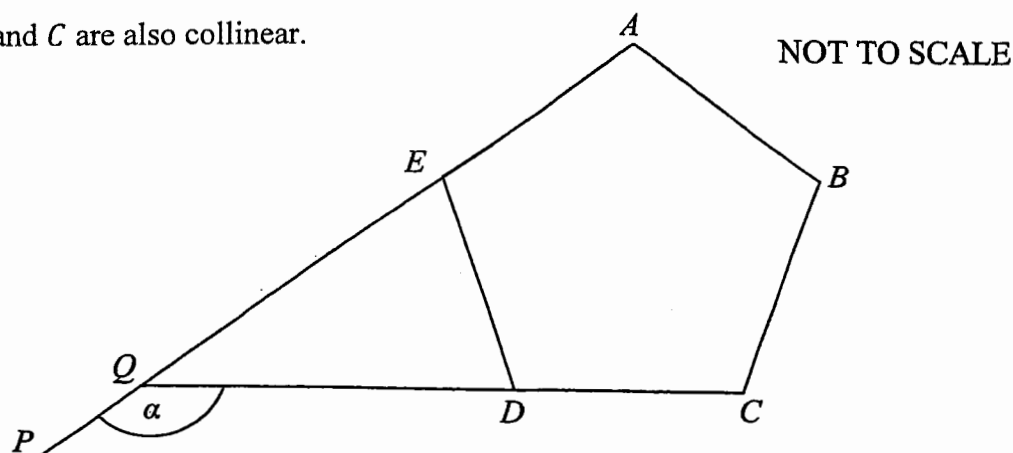
Question 2 (8 marks) Start a new page

- a) Solve $x^2 = 5x$. 2
- b) Factorise $3x^2 + x - 2$. 1
- c) Find the exact value of θ such that $2\cos\theta = 1$ if $0 \leq \theta \leq 90^\circ$. 2
- d) i) Sketch $y = 9 - x^2$ showing x and y intercepts. 2
- ii) Find the range of $y = 9 - x^2$. 1

Question 3 (8 marks) Start a new page

- a) Let M be the midpoint of $(-1,4)$ and $(5,8)$. 3
- Find the equation of the line through M with gradient $-\frac{1}{2}$ (write your answer in general form).
- b) Prove that $\frac{\sin\theta}{1-\cos\theta} + \frac{\sin\theta}{1+\cos\theta} = 2\operatorname{cosec}\theta$. 3

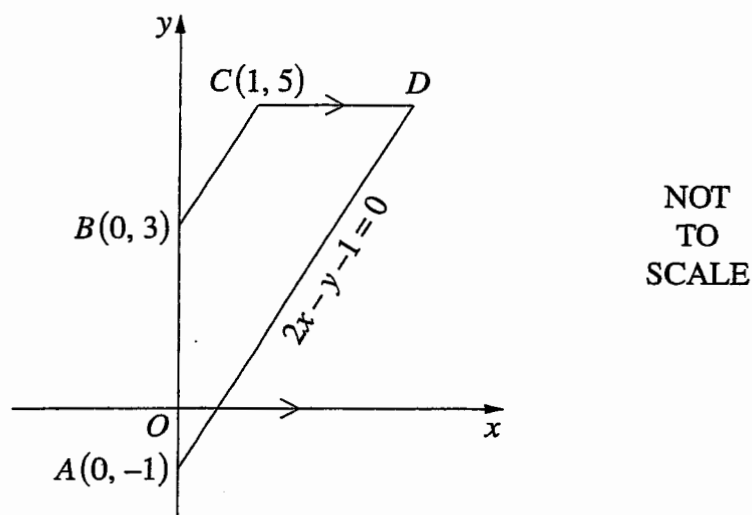
- c) $ABCDE$ is a regular pentagon. The points P, Q, E and A are collinear. The points Q, D and C are also collinear.



Find the size of angle a giving reasons.

2

Question 4 (8 marks) Start a new page



In the diagram, $ABCD$ is a quadrilateral. The equation of the line AD is $2x - y - 1 = 0$.

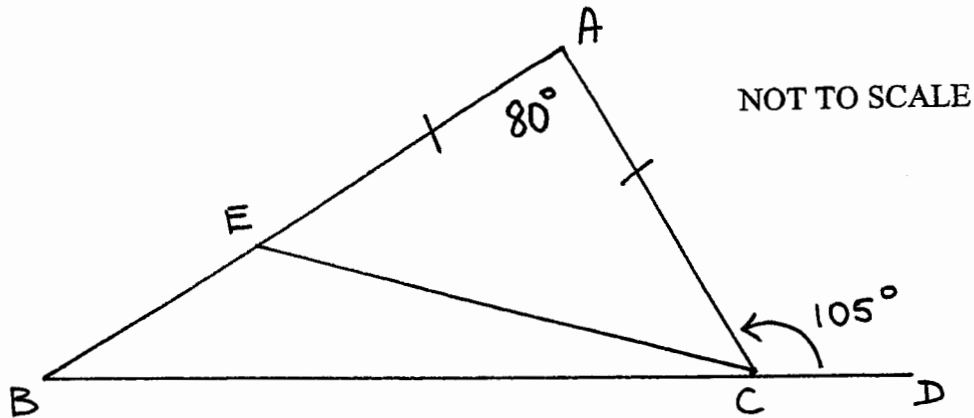
- Show that $ABCD$ is a trapezium by showing that BC is parallel to AD . 2
- The line CD is parallel to the x -axis. Find the coordinates of D . 1
- Find the length of BC . 1
- Show that the perpendicular distance from B to AD is $\frac{4}{\sqrt{5}}$. 2
- Hence, or otherwise, find the area of the trapezium $ABCD$.
(simplify your answer fully) 2

Question 5 (8 marks) Start a new page

a) If $\sin(3x - 20^\circ) = \cos(2x + 50^\circ)$ find x .

1

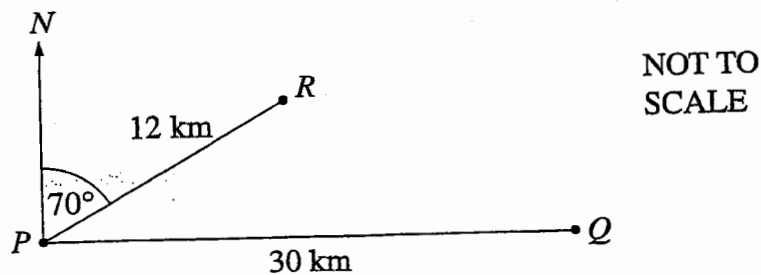
b) In the diagram below, $AE = AC$, $\hat{EAC} = 80^\circ$ and $\hat{ACD} = 105^\circ$.



3

Prove, with full reasoning, that $EB = EC$

c)



The diagram shows a point P which is 30 km due west of the point Q .

The point R is 12 km from P and has a bearing from P of 070° .

i) Find the distance of R from Q (to nearest km).

2

ii) Find the bearing of R from Q (to nearest degree).

2

Question 6 (8 marks) Start a new page

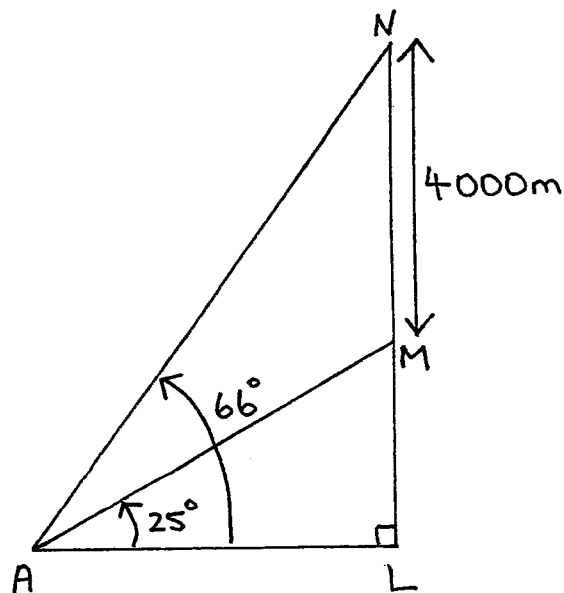
- a) Find x and y such that $x + y - \sqrt{x - y} = 18 - 2\sqrt{3}$. 2
- b) Solve the simultaneous equations
 $4x - y = 9$
 $3xy = -6$ 2
- c) i) Sketch the function defined as follows
$$f(x) = \begin{cases} 0 & \text{if } x \leq -3 \\ -1 & \text{if } -3 < x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$$
 3
- ii) Find $f(a^2)$ 1

Question 7 (8 marks) Start a new page

- a) Solve $5x^2 - 12x + 4 < 0$. 2
- b) If $\sin\theta = \frac{-1}{\sqrt{13}}$ and $\tan\theta > 0$ find the exact value of $\cos\theta$. 2
- c) Solve in the domain $0 \leq \theta \leq 360^\circ$
- i) $2\cos^2\theta = 1$ 2
- ii) $(\sin\theta - 1)(2\sin\theta - 1) = 0$ 2

Question 8 (8 marks) Start a new page

- a) A rocket launched vertically from L is observed from A . Soon after launch when at position M its angle of elevation is 25° . After it climbs 4000 metres from this position to N its angle of elevation is 66° .



- i) Find $\hat{ANL}$ and $\hat{NAM}$. 2
 - ii) Find AM to the nearest metre. 2
 - iii) Find how far the observer is, horizontally from the launch pad.
(answer to nearest metre) 2
- b) Solve $3\sin\theta = 2\cos\theta$ if $-180^\circ \leq \theta \leq 180^\circ$.
Write your solutions correct to the nearest minute. 2

Question 1

a) 6.90 (3 sig. fig)

b) $\frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} = \frac{\sqrt{5}+2}{1}$
 $\therefore a=2 \quad b=1$

c) $(x+3)^2 + (y-4)^2 = 36$

d) $2x+3=9 \quad 2x+3=-9$
 $2x=6 \quad 2x=-12$
 $x=3$ and $x=-6$

e) $6-x \geq 0$
 $-x \geq -6$

$D: x \leq 6$

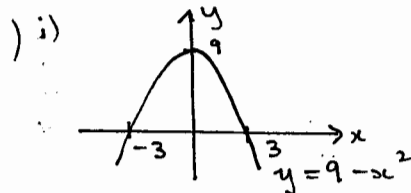
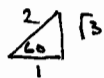
Question 2

a) $x^2-5x=0$
 $x(x-5)=0 \quad \therefore x=0, x=5$

b) $3x^2+x-2$
 $3x \times -2$
 $3x \times -1$
 $3x \times -1$
 $3x \times -1$

$(3x-2)(x+1)$

c) $2 \cos \theta = 1$
 $\cos \theta = \frac{1}{2}$
 $\therefore \theta = 60^\circ$



ii) R: $y \leq 9$

Question 3

a) $M(2,6)$
 $y-6 = -\frac{1}{2}(x-2)$
 $2y-12 = -x+2$
 $x+2y-14=0$

b) $LHS = \frac{\sin \theta}{1-\cos \theta} + \frac{\sin \theta}{1+\cos \theta}$
 $= \frac{\sin \theta(1+\cos \theta) + \sin \theta(1-\cos \theta)}{(1-\cos \theta)(1+\cos \theta)}$
 $= \frac{\sin \theta + \sin \theta \cos \theta + \sin \theta - \sin \theta \cos \theta}{1-\cos^2 \theta}$
 $= \frac{2 \sin \theta}{\sin^2 \theta}$
 $= \frac{2}{\sin \theta}$
 $= 2 \operatorname{cosec} \theta$
 $= RHS$

c) $\hat{OED} = \hat{ODE} = 72^\circ$ (ext. angle of regular pentagon)
 $\therefore \alpha = 144^\circ$ (angle sum of triangle)

Question 4

a) $2x-y-1=0$
 $\therefore y=2x-1 \quad m_{AB}=2$
 $m_{BC} = \frac{5-3}{1-0} = 2$
 $\therefore m_{AD} = m_{BC} \therefore AD \parallel BC$

b) $D(3,5)$

c) $BC = \sqrt{(1-0)^2 + (5-3)^2}$
 $= \sqrt{1+4}$
 $= \sqrt{5}$ units

d) $P = \left| \frac{2x_0 - 1 \times 3 - 1}{\sqrt{2^2 + 1^2}} \right|$
 $= \left| \frac{-4}{\sqrt{5}} \right|$

$\therefore P = \frac{4}{\sqrt{5}}$

e) $\text{Area } ABCD = \frac{1}{2} \cdot \frac{4}{\sqrt{5}} (BC + AD)$

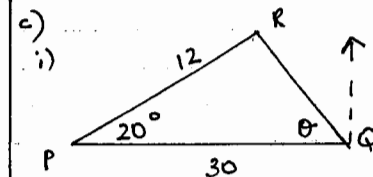
$BC = \sqrt{5}$ from above
 $AD = \sqrt{(3-0)^2 + (5-1)^2}$
 $= \sqrt{9+16}$
 $= \sqrt{25}$
 $= 5$

$\therefore \text{Area} = \frac{1}{2} \cdot \frac{4}{\sqrt{5}} (\sqrt{5} + 5)$
 $= \frac{1}{2} \cdot \frac{4}{\sqrt{5}} (4\sqrt{5})$
 $= 8 \text{ unit}^2$

Question 5

a) $3x-20 + (2x+50) = 90$
 $5x + 30 = 90$
 $5x = 60$
 $x = 12^\circ$

b) $\hat{EBC} = 25^\circ$ (ext. angle of triangle)
 $\hat{ACE} = 50^\circ$ (opp equal sides isos. triangle)
 $\therefore \hat{ECB} = 25^\circ$ (angles on straight line)
 $\therefore EB = EC$ (opp equal angles in isosceles triangle)



$RQ^2 = 12^2 + 30^2 - 2 \times 12 \times 30 \times \cos 20^\circ$
 $RQ = 19 \text{ km}$ (to nearest km)

ii) $\frac{\sin \theta}{12} = \frac{\sin 20^\circ}{19}$
 $\sin \theta = \frac{12 \sin 20^\circ}{19}$

$\theta = 12.47^\circ$

$\theta = 12^\circ$ to nearest deg.

bearing R from Q
 $N 78^\circ W$ OR $282^\circ T$
 (if using cosine Rule and $19 \text{ km} \rightarrow 282^\circ T$)

Question 6

a) $x+y=18$
 $x-y=12$

$2x=30$

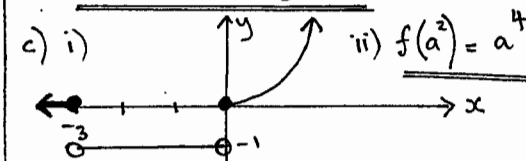
$\therefore x=15 \quad y=3$

b) $4x-y=9$
 $\therefore y=4x-9$
 $3x(4x-9)=-6$
 $12x^2-27x+6=0$
 $4x^2-9x+2=0$

$4x \times -1$
 $3x \times -2$

$(4x-1)(x-2)=0$

$\therefore \begin{cases} x = \frac{1}{4} \text{ and } x = 2 \\ y = -8 \text{ and } y = -1 \end{cases}$

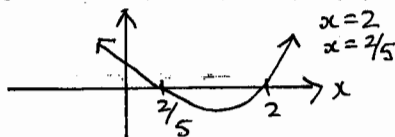


Question 7

a) $5x^2 - 12x + 4 < 0$

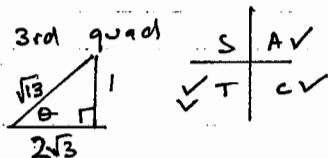
$5x \begin{matrix} \times -2 \\ \times -2 \end{matrix}$

$(5x - 2)(x - 2) < 0$



$\frac{2}{5} < x < 2$

b) θ in 3rd quad



$\therefore \cos \theta = -\frac{2\sqrt{3}}{\sqrt{13}}$

c)

i) $2\cos^2 \theta = 1$

$\cos^2 \theta = \frac{1}{2}$

$\cos \theta = \pm \frac{1}{\sqrt{2}}$

acute $\theta = 45^\circ$

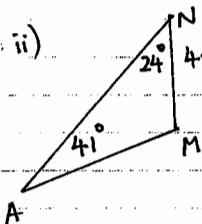
$\therefore \theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

ii) $\sin \theta = 1$ & $\sin \theta = \frac{1}{2}$ ✓
acute $\theta = 30^\circ$ ✓

$\therefore \theta = 90^\circ$ and $\theta = 30^\circ, 150^\circ$

Question 8

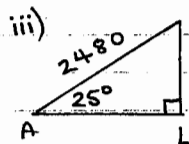
a) i) $\hat{ANL} = 24^\circ$
 $\hat{NAM} = 41^\circ$



$\frac{AM}{\sin 24^\circ} = \frac{4000}{\sin 41^\circ}$

$AM = \frac{4000 \sin 24^\circ}{\sin 41^\circ}$

$AM = 2480$ to nearest m



$\cos 25^\circ = \frac{AL}{2480}$

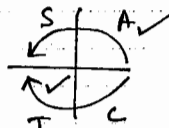
$AL = 2247.6$ m

$\therefore AL = 2248$ m (nearest metric)

b) $3\sin \theta = 2\cos \theta$

$\frac{\sin \theta}{\cos \theta} = \frac{2}{3}$

$\therefore \tan \theta = \frac{2}{3}$



$\therefore$ acute angle $33^\circ 41'$

$\therefore \theta = 33^\circ 41', -146^\circ 19'$